

# Laboratory of Computational Fluid Dynamics

## Exercise Report #3

### 2D Incompressible Navier-Stokes Solver

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## 1 Introduction and Governing Equations

This report details the development, verification, and analysis of a numerical code designed to solve the 2D incompressible Navier-Stokes equations within a rectangular domain. The governing equations for a constant-density fluid are expressed in their non-dimensional form as follows:

$$\begin{cases} \frac{\partial u_i}{\partial t} = -\frac{\partial}{\partial x_j}(u_i u_j) - \frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j \partial x_j} \\ \frac{\partial u_i}{\partial x_i} = 0 \end{cases} \quad (1)$$

where  $u$  and  $v$  denote the horizontal and vertical velocity components,  $\nu$  is the kinematic viscosity, and  $\rho$  is the density.

## 2 Numerical Implementation

### 2.1 Starting Point

The solver was not written entirely from scratch. The professor provided a Fortran code template implementing the overall architecture of a fractional-step, projection-based solver for the two-dimensional incompressible Navier-Stokes equations, discretized in space with second-order finite differences on a staggered grid and integrated in time with a three-stage Runge-Kutta (RK3) scheme. The template already contained the module layout, the data structures, the boundary-condition handling, the Poisson solver, and the input/output routines. Several core numerical routines, however, were deliberately left incomplete (marked with placeholder `...` statements) and had to be implemented in order to obtain a working solver. The present section documents the code structure inherited from the template, the routines that were completed, and a numerical issue that was encountered and resolved during validation.

### 2.2 Code Structure

The code is organized as a collection of Fortran modules, each responsible for a well-defined part of the solution algorithm. The source files are:

- `types.f90` – defines the working precision (`rp`, double precision) used consistently throughout the code.
- `common.f90` – defines the global state of the simulation: the velocity and pressure arrays, the grid coordinates and spacings, the physical parameters ( $\nu$ ,  $\rho$ ), the time-stepping parameters, and the boundary-condition flags/values. It also provides the grid initialization and field allocation routines.
- `bc.f90` – applies boundary conditions (Dirichlet, Neumann, periodic) to the velocity and pressure ghost cells on all four sides of the domain, consistently with the staggered arrangement of the variables.
- `rhs.f90` – evaluates the right-hand side of the momentum equations: the convective term (in conservative/skew-symmetric form), the diffusive term (second-order centered Laplacian), and the pressure gradient term.
- `pressure_correction.f90` – computes the divergence of an intermediate (non-solenoidal) velocity field and applies the pressure correction step of the projection method to enforce a divergence-free velocity at the end of each Runge-Kutta stage.
- `poisson.f90` – solves the discrete Poisson equation for the pressure correction using a preconditioned BiCGStab iterative solver, including the boundary-condition treatment of the pressure unknown (Dirichlet, Neumann, periodic) and the gauge-fixing required when only Neumann/periodic conditions are present.

- `timeint.f90` – assembles the previous building blocks into the three-stage fractional-step RK3 time advancement: at each stage it evaluates the momentum right-hand side, forms the intermediate (predictor) velocity, projects it onto a divergence-free space by calling the Poisson solver and the pressure-correction routine, and updates the pressure field.
- `io.f90` – handles diagnostic output to screen (CFL numbers, divergence norms, Poisson convergence), and writes the solution fields to disk in binary (restart) and ASCII/VTK format for post-processing.
- `main.f90` – the driver program: it initializes the grid, the physical and numerical parameters, the boundary conditions, the initial fields, and runs the time-stepping loop, calling `TIMEINT_AdvanceRK3FractionalStep` at every step. Since many tasks have to be studied, the name of the `main.f90` to be run is left unchanged, while the others main code files have the name temporarily modified. The name of the data file obtained is then modified before re-running the code in order not to overwrite it.

During execution, the code is structured to provide a real-time diagnostic log via terminal output. This continuous feedback is a fundamental element for monitoring numerical stability and the overall health of the simulation. Specifically, at each save interval, the solver calculates and reports the CFL number, allowing the user to verify that the combination of the spatial grid size and the *timestep*  $\Delta t$  strictly complies with the convective and viscous stability limits imposed by the explicit scheme. Furthermore, the terminal outputs the number of iterations required by the Poisson's equation solver and the magnitude of the divergence residual, providing immediate quantitative feedback on the correct conservation of mass at every stage of the time integration.

## 2.3 Routines Completed

The following routines were incomplete in the provided template and were implemented as part of this work.

### 2.3.1 Momentum right-hand side (`rhs.f90`)

The pressure-gradient contribution to the momentum equation was completed as

```
1 rhs_u_out(i,j) = -(p(i+1,j) - p(i,j)) / (dx * rho)
2 rhs_v_out(i,j) = -(p(i,j+1) - p(i,j)) / (dy * rho)
```

while the convective and diffusive contributions were discretized with standard second-order, conservative finite-difference stencils on the staggered grid:

```
1 duu_dx = ( ((u(i+1,j)+u(i,j))*0.5_rp)**2 - &
2           ((u(i,j)+u(i-1,j))*0.5_rp)**2 ) / dx
3 d2u_dx2 = (u(i+1,j) - 2.0_rp*u(i,j) + u(i-1,j)) / (dx*dx)
4 d2u_dy2 = (u(i,j+1) - 2.0_rp*u(i,j) + u(i,j-1)) / (dy*dy)
```

and analogously for the convective/diffusive terms of the *v*-momentum equation.

### 2.3.2 Pressure correction (`pressure_correction.f90`)

The divergence of the predictor velocity field and the velocity-correction step were completed as

```
1 div(i,j) = (velx(i,j) - velx(i-1,j)) / dx + &
2           (vely(i,j) - vely(i,j-1)) / dy

1 u(i,j) = ustar(i,j) - dt_rk3 * (p_star(i+1,j) - p_star(i,j)) / dx
2 v(i,j) = vstar(i,j) - dt_rk3 * (p_star(i,j+1) - p_star(i,j)) / dy
```

### 2.3.3 Time integration (`timeint.f90`)

The predictor step for the  $v$ -velocity (analogous to the one already provided for  $u$ ) and the call to the projection routine inside each RK3 stage were completed:

```
1 vstar(i,j) = v_inout(i,j) + fct_3 * rhs_v(i,j)
2 vstar(i,j) = vstar(i,j) + fct_1 * rhs_v(i,j) + fct_2 * rhss_v(i,j)
3 ...
4 call ProjectStageVelocity(u_inout, v_inout, ustar, vstar, p_star, dt_rk3)
```

together with the two missing calls inside `ProjectStageVelocity`, which apply the pressure boundary conditions and the velocity correction:

```
1 call BC_ApplyPressure(pstage)
2 call PC_CorrectVelocity(uout, vout, uin, vin, pstage, dt_stage)
```

## 2.4 A Note on the Staggered-Grid Indexing Convention

During validation on the Poiseuille test case, the predictor-velocity divergence grew exponentially within a few time steps until the solution diverged, even though the Poisson solver converged to machine precision at every stage and the time step satisfied the usual stability limits. The cause was an inconsistent indexing convention for the pressure gradient: on a staggered grid,  $u(i, j)$  can be associated with either the left face of cell  $i$ , giving  $(p(i, j) - p(i - 1, j))/\Delta x$ , or the right face, giving  $(p(i + 1, j) - p(i, j))/\Delta x$  – both valid second-order choices, but the momentum right-hand side and the projection step must use the *same* one.

In an early version, `rhs.f90` used the right-face convention while in `pressure_correction.f90` I (wrongly) used the left-face one; this half-cell mismatch meant the projection corrected for a different operator than the one that generated the divergence, amplifying it instead of removing it at every stage. Once both routines were made consistent, the predictor divergence remained bounded and the simulation was stable.

This illustrates a general requirement of staggered-grid projection methods: the gradient, divergence, and Laplacian operators used in the predictor, the Poisson equation, and the correction step must be mutually consistent discretizations of the same continuous operators.

## 3 Task 1: Validation of Diffusive Terms – Poiseuille Flow

The primary objective of this task is to validate the accurate implementation of the diffusive terms and the treatment of physical boundaries by simulating a standard 2D Poiseuille flow.

### 3.1 Computational Setup

The simulation is performed on a rectangular domain of size  $L_x \times L_y = 10 \times 1$ , discretized with a grid of  $N_x \times N_y = 100 \times 40$  nodes. The flow is initialized with zero velocity and pressure fields. The boundary conditions are applied as follows:

- **Bottom and Top Walls** ( $y = 0$  and  $y = L_y$ ): No-slip condition ( $u = 0, v = 0$ ) and zero pressure gradient ( $\partial p / \partial y = 0$ ).
- **Inlet** ( $x = 0$ ): Uniform inflow velocity profile  $u(y) = 1$  with zero pressure gradient ( $\partial p / \partial x = 0$ ).
- **Outlet** ( $x = L_x$ ): Reference Dirichlet pressure  $p = 0$  and zero velocity gradients ( $\partial u / \partial x = 0, \partial v / \partial x = 0$ ).

The fluid density is set to  $\rho = 1$ , and the viscosity is derived from the bulk Reynolds number  $Re = U_b h / \nu = 500$ , considering  $U_b = 1$  and  $h = 1$ . The integration spans  $t \in [0, 20]$  using a time step  $\Delta t = 10^{-2}$  to guarantee stability. The solution fields are stored at equi-spaced time intervals of 1 second.

## 3.2 Results and Discussion

### Qualitative validation

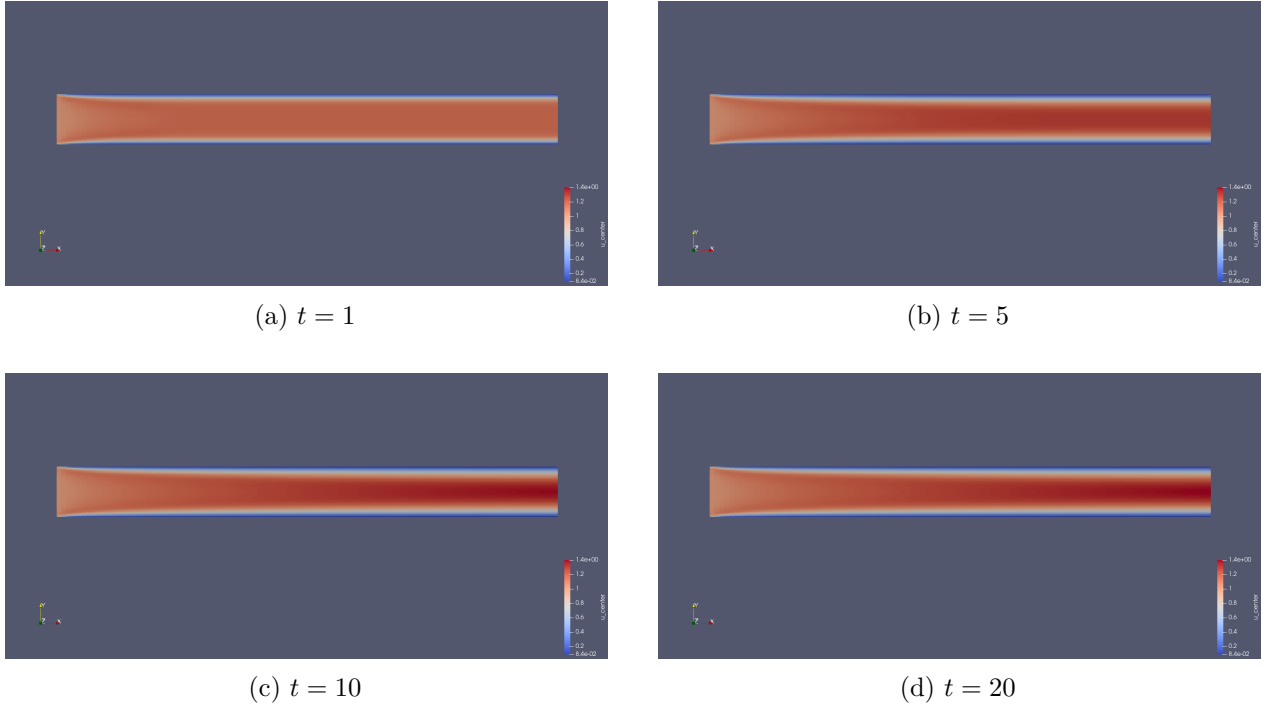


Figure 1: Temporal evolution of the horizontal velocity field ( $u$ ).

Figure 1 shows that the temporal evolution of the horizontal velocity field ( $u$ ) qualitatively demonstrates the expected physical behavior for a developing channel flow.

The fluid enters the domain with a uniform velocity profile and, due to the no-slip condition at the top and bottom walls, viscous boundary layers begin to grow (this characteristic is highlighted by the comparison between figures 1a and 1b).

To satisfy mass conservation, the deceleration of the fluid near the boundaries forces the core region of the flow to accelerate (this characteristic is highlighted by the change in the color scale of the red core region going from figure 1a to 1c).

By comparing figure 1c with figure 1d, we see that the flow visually reaches a fully developed state, showing no further significant gradients along the streamwise direction.

While these contour maps confirm that the solver correctly captures the macroscopic flow dynamics, a rigorous quantitative validation will be performed in the next section by comparing the numerical profile against the exact analytical Poiseuille parabola.

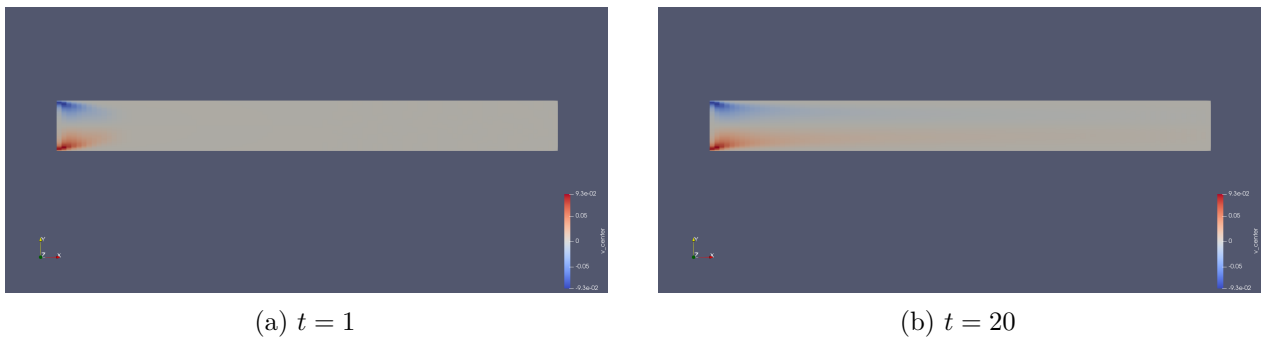


Figure 2: Temporal evolution of the vertical velocity field ( $v$ ).

The vertical velocity field ( $v$ ) at  $t = 1$  and  $t = 20$  clearly illustrates the kinematic adjustment required at the channel inlet. While the theoretical Poiseuille flow assumes a strictly zero vertical velocity everywhere, the transition from a uniform flat profile to a parabolic one forces the fluid to migrate from the walls towards the centerline to satisfy mass conservation. This phenomenon is captured by the symmetric positive (red) and negative (blue) velocity lobes near the inlet. As the flow reaches a steady state at  $t = 20$ , this developing region stabilizes, and the vertical velocity correctly dissipates to zero along the rest of the channel, in full qualitative agreement with the one-dimensional analytical assumption.

## Quantitative validation

To quantitatively evaluate the transient behavior of the flow, the horizontal velocity profile  $u(y)$  is extracted at the cross-section  $x = 9.5$  for different simulation times. Figure 3 displays the progressive development of the velocity field compared to the exact analytical solution for a fully developed Poiseuille flow,  $u(y) = 6y(1 - y)$ .

At  $t = 1$ , the profile is still largely uniform in the core region, with steep velocity gradients confined to the viscous boundary layers near the upper and lower walls. As time progresses ( $t = 5$ ), the boundary layers grow thicker, forcing the central inviscid core to accelerate to satisfy mass conservation.

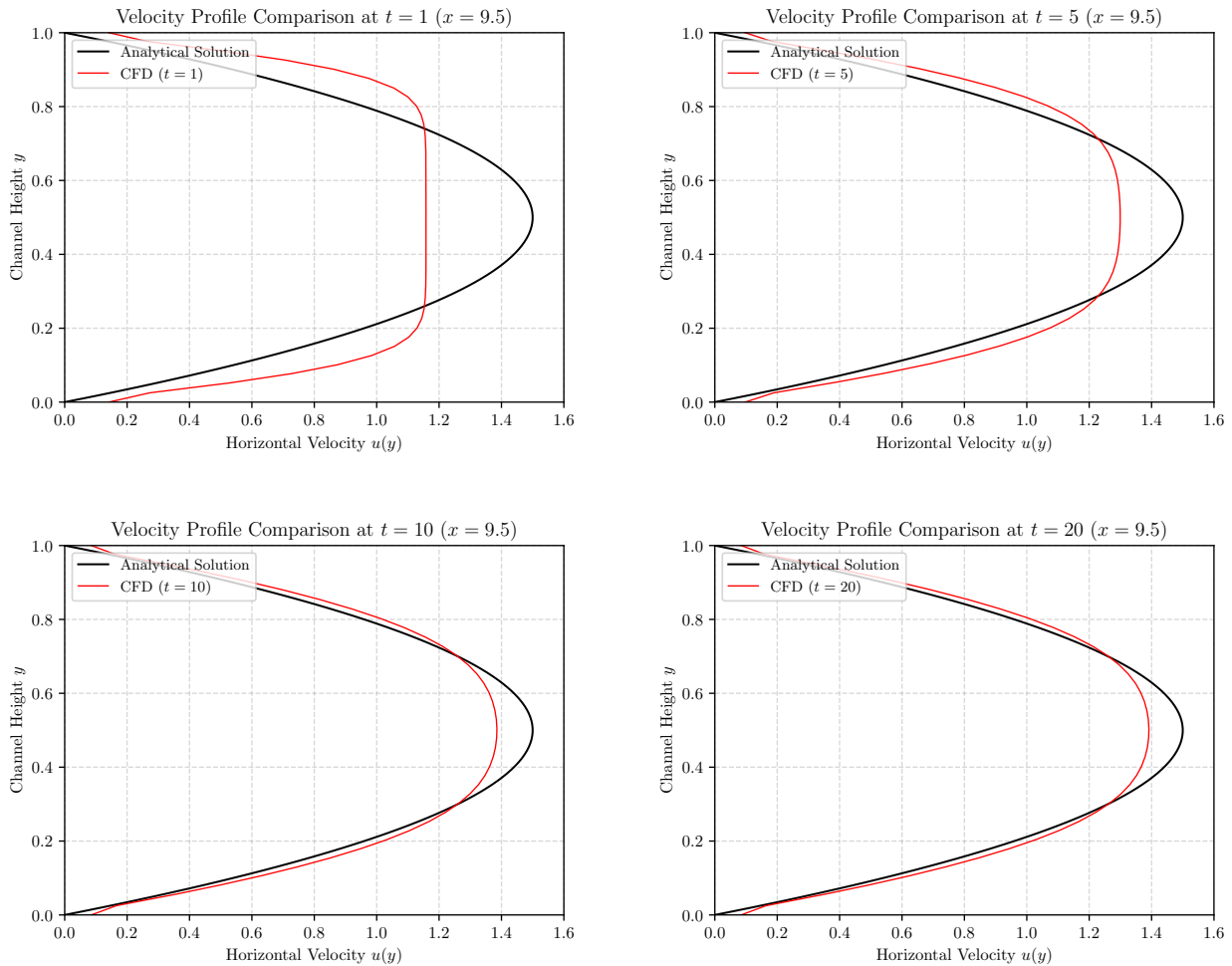


Figure 3: Individual velocity profiles at  $x = 9.5$  for discrete time steps, plotted against the analytical fully developed solution.

A comprehensive view of the temporal evolution and a detailed inspection of the velocity peak are provided in Figure 4. A critical observation can be made regarding the solver's convergence: the

numerical curves for  $t = 10$  and  $t = 20$  nearly overlap across the entire channel height.

To better appreciate the asymptotic behavior of the fractional-step scheme, Figure 4(b) offers a zoomed-in view focusing exclusively on the vertex of the parabola ( $0.35 \leq y \leq 0.65$ ). This close-up inspection clearly illustrates that the residual acceleration between  $t = 10$  and  $t = 20$  is minimal, confirming that the transient phase has effectively exhausted and the simulation has achieved a rigorous steady-state condition.

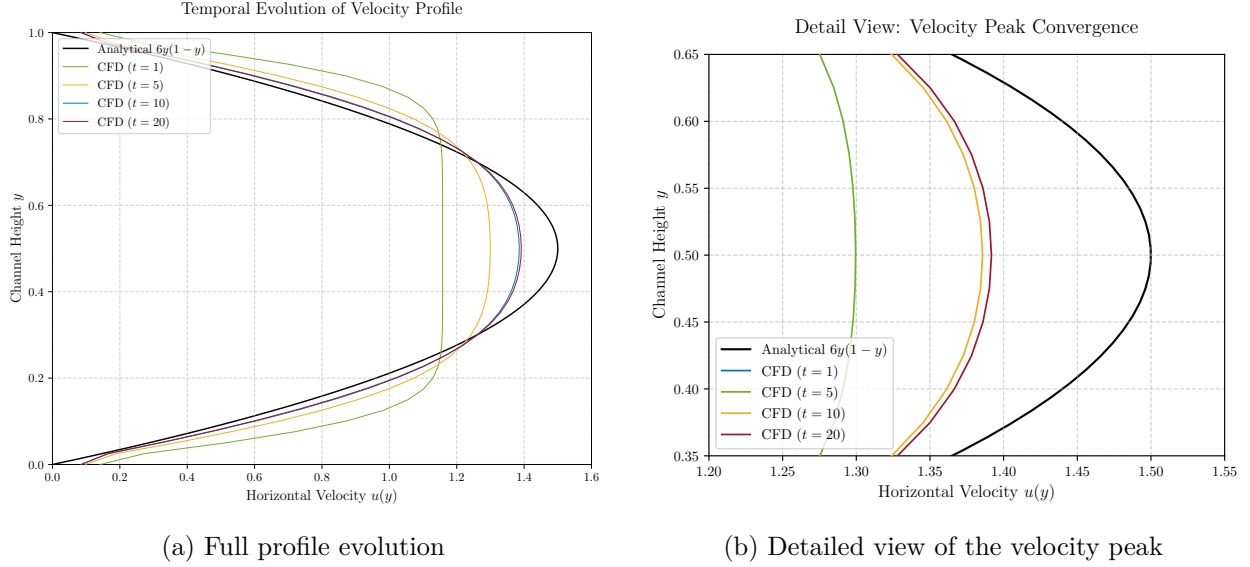


Figure 4: Combined temporal evolution of the velocity profile at  $x = 9.5$ : (a) overview of the developing flow, and (b) zoomed detail highlighting the asymptotic convergence near the centerline.

However, as evidenced by both views, the steady-state CFD profile at  $x = 9.5$  does not perfectly match the analytical solution, exhibiting a slightly flatter core (with a lower peak velocity around  $u \approx 1.4$  instead of 1.5) and fuller edges. Rather than a numerical error, this discrepancy represents a correct physical feature captured by the solver: the flow at  $x = 9.5$  has not yet achieved a fully developed state.

According to hydrodynamic theory, the entrance length  $L_e$  required for the boundary layers to merge at the centerline is proportional to the Reynolds number ( $L_e \approx 0.06 \cdot Re \cdot D_h$ ). For the domain and Reynolds number configured in this simulation, the section  $x = 9.5$  still falls within the development region. Consequently, the core flow is still gradually accelerating, correctly justifying the deviation from the asymptotic analytical solution.

## 4 Task 2: Validation – Taylor-Green Vortex Benchmark

The Taylor-Green vortex constitutes an ideal benchmark to assess the solver's numerical accuracy, its capacity to conserve kinetic energy, and the strict enforcement of fully periodic boundary conditions.

### 4.1 Computational Setup

The domain is defined as a square of dimensions  $L_x \times L_y = 2\pi \times 2\pi$  and discretized with  $64 \times 64$  grid nodes. Fully periodic boundary conditions are enforced on all four boundaries.

The flow is initialized with the analytical divergence-free conditions:

$$u(x, y) = U_0 \sin(kx) \cos(ky) \quad (2)$$

$$v(x, y) = -U_0 \cos(kx) \sin(ky) \quad (3)$$

$$p(x, y) = \frac{\rho U_0^2}{4} [\cos(2kx) + \cos(2ky)] \quad (4)$$

Setting  $U_0 = 1$ ,  $k = 1$ ,  $\rho = 1$ , and  $\nu = 10^{-2}$ , the problem is advanced in time up to  $t = 75$  with  $\Delta t = 10^{-2}$ .

## 4.2 Results and Discussion

### Qualitative validation

Figure 5 presents the velocity field at four distinct computational times:  $t = 0, 25, 50$ , and  $75$  s. The visualizations combine velocity magnitude contours with seeded streamlines and directional vectors (glyphs) to illustrate both the energy dissipation and the kinematic topology of the flow.

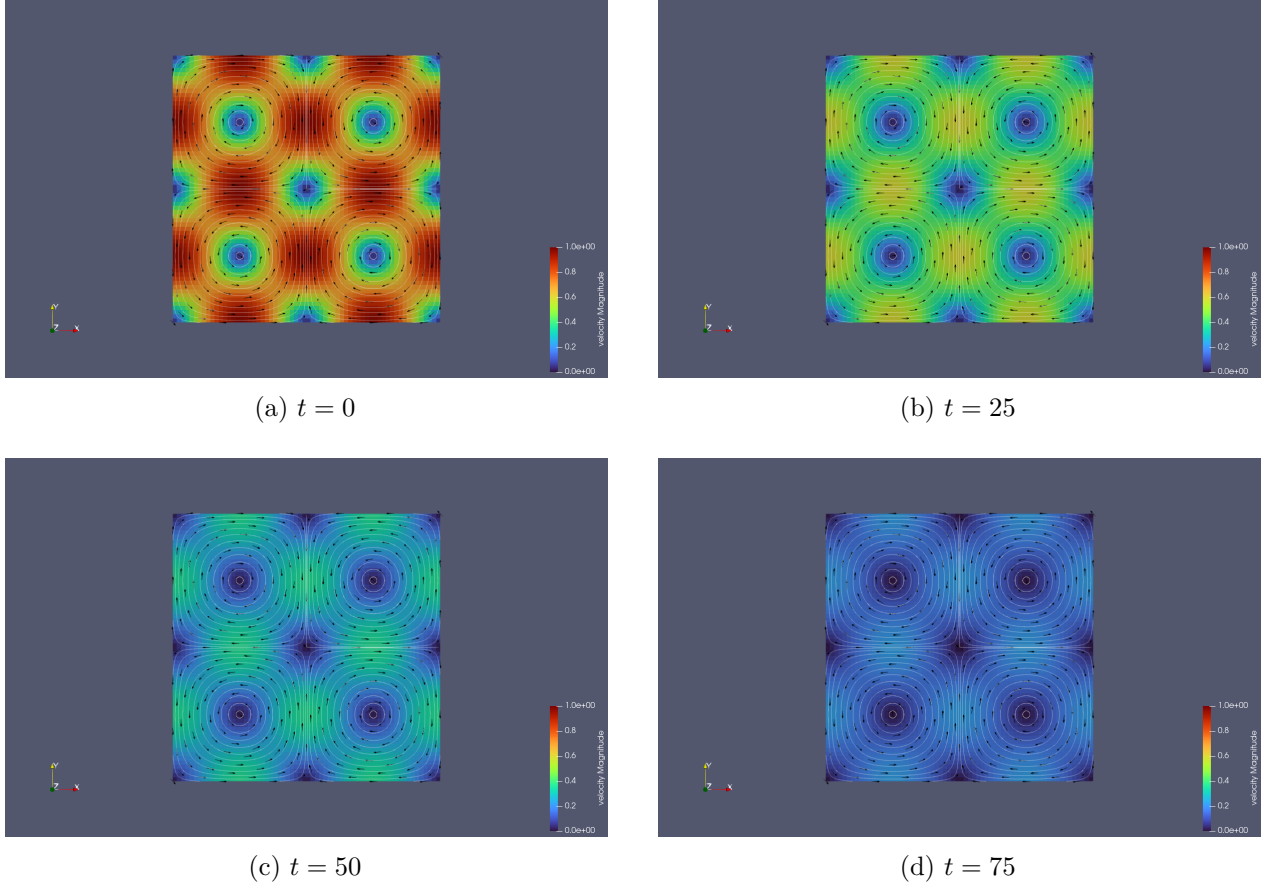


Figure 5: Velocity magnitude contours, streamlines, and velocity vectors of the Taylor-Green vortex. A fixed global color scale (0.0 to 1.0) is maintained across all frames to visualize the exponential kinetic energy decay.

This visual strategy effectively highlights the purely viscous decay of the kinetic energy inherent to this unforced system. At  $t = 0$ , the peak velocities at the vortex peripheries reach the maximum initialized value ( $|\mathbf{V}| \approx 1.0$ , indicated by the red regions). As time progresses, the action of the kinematic viscosity ( $\nu = 10^{-2}$ ) smoothly dampens the velocity field, gradually shifting the global contour map towards the lower end of the spectrum (dark blue regions) by  $t = 75$  s.

Crucially, while the velocity magnitude decays exponentially, the spatial topology of the flow remains perfectly intact. The streamlines demonstrate that the geometric centers of the vortices and the orthogonal stagnation lines separating them do not drift, merge, or deform. The continuous preservation of these symmetric, gear-like vortical structures serves as strong evidence that the adopted spatial discretization scheme correctly models the flow without introducing asymmetric numerical dispersion or artificial deformations.

## Quantitative Validation

To rigorously assess the solver's accuracy, a quantitative comparison is performed along the horizontal centerline of the domain ( $y = \pi$ ) at  $t = 25, 50$ , and  $75$  seconds. The numerical results extracted from the CFD simulations are verified against the exact analytical expressions for the horizontal velocity component  $u(x, y, t)$  and the pressure field  $p(x, y, t)$ , given respectively by:

$$u(x, y, t) = U_0 \sin(kx) \cos(ky) e^{-2\nu k^2 t} \quad (5)$$

$$p(x, y, t) = \frac{\rho U_0^2}{4} [\cos(2kx) + \cos(2ky)] e^{-4\nu k^2 t} \quad (6)$$

Setting the benchmark parameters ( $U_0 = 1$ ,  $k = 1$ ,  $\rho = 1$ , and  $\nu = 10^{-2}$ ), the profiles evaluated at the centerline simplify to  $u(x, \pi, t) = -\sin(x)e^{-0.02t}$  and  $p(x, \pi, t) = \frac{1}{4}(\cos(2x) + 1)e^{-0.04t}$ . Because the absolute pressure level is physically arbitrary in incompressible flows, the numerical pressure data has been shifted to plot the relative pressure  $\Delta p$ , aligning the reference minimum with the analytical exact zero.

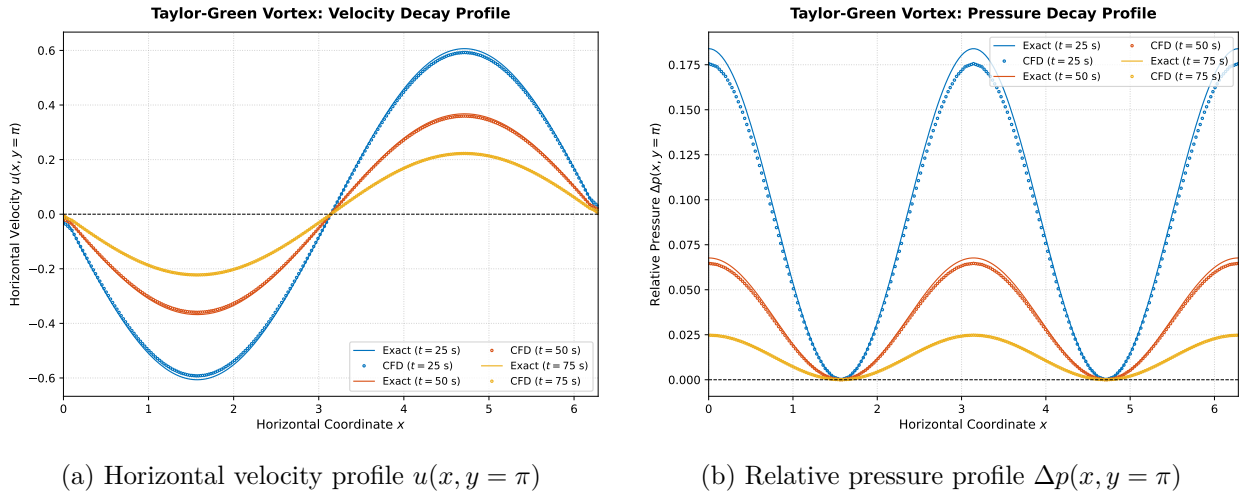


Figure 6: Comparison between the analytical exact solution (solid lines) and the computed CFD data (discrete markers) along the horizontal centerline. The solver accurately captures the spatial frequencies and the exponential decay trends.

The quantitative results shown in Figure 6 confirm that the numerical code matches the analytical framework with high precision. For both velocity and pressure, the numerical markers exhibit perfect phase alignment with the exact sinusoidal curves. The complete absence of phase shift (numerical dispersion) confirms that the central difference discretization perfectly preserves the spatial symmetries of the periodic domain without introducing artificial wave transport.

A detailed inspection of the crests reveals a slight underprediction of the peak amplitudes, which is most evident in the early stages of the decay (e.g.,  $t = 25$  s). This attenuation is a well-known manifestation of **numerical dissipation** (truncation error), acting as an artificial viscosity that causes the kinetic energy to decay slightly faster than the purely physical viscous limit.

Notably, this amplitude discrepancy appears significantly more pronounced in the pressure field than in the velocity field. This behavior is physically consistent: in incompressible Navier-Stokes formulations, pressure is governed by a Poisson equation driven by the velocity gradients, meaning that pressure scales quadratically with velocity ( $p \propto U^2$ ). Consequently, any small numerical damping introduced by the truncation error in the velocity field is quadratically amplified when computing the corresponding pressure field. Overall, the agreement remains highly satisfactory, and further reduction of this amplitude gap could be monotonically achieved through finer spatial grid resolution.

## 5 Task 3: Free Simulation – Kelvin-Helmholtz Instability

To evaluate the solver’s capability to handle non-linear flow evolution, high-gradient regions, and vortex dynamics, the Kelvin-Helmholtz (K-H) instability was selected as the final test case. This phenomenon occurs when a velocity shear is present within a continuous fluid, representing a fundamental mechanism in turbulent mixing layers and aerodynamic wakes.

### 5.1 Computational Setup

The computational domain is defined as a rectangle of dimensions  $L_x \times L_y = 2.0 \times 1.0$ , discretized with a  $128 \times 64$  Cartesian grid. To effectively isolate the shear layer at the center of the domain, a specific combination of boundary conditions is applied:

- **East/West ( $x$ -direction):** Fully periodic boundaries to allow the continuous propagation of the spatial instability.
- **North/South ( $y$ -direction):** Impermeable moving walls (Dirichlet for velocity, Neumann for pressure). The upper wall moves at  $U_{north} = U_0$  while the lower wall moves at  $U_{south} = -U_0$ , continuously driving the shear flow.

The flow is initialized with a smoothed top-hat profile using a hyperbolic tangent function to prevent infinite numerical gradients at  $t = 0$ , coupled with a small, localized vertical perturbation to trigger the instability:

$$u(x, y) = U_0 \tanh\left(\frac{y - 0.5}{\delta_0}\right) \quad (7)$$

$$v(x, y) = A \sin(k_x x) \exp\left(-\frac{(y - 0.5)^2}{2\delta_0^2}\right) \quad (8)$$

The selected wavenumber  $k_x = 2\pi$  implies exactly one complete wavelength over the streamwise domain length  $L_x = 2.0$ , since  $\lambda = 2\pi/k_x = 1.0 = L_x/2$ . This configuration excites the fundamental mode of the periodic domain, which is known to be the most unstable one according to linear stability theory. As a direct consequence, the nonlinear saturation of this single-mode perturbation naturally produces **two** counter-rotating vortices symmetrically arranged along the shear layer, one per half-wavelength of the domain.

The simulation is executed with  $U_0 = 1.0$ , a shear layer thickness  $\delta_0 = 0.03$ , perturbation amplitude  $A = 0.01$ , and wave number  $k_x = 2\pi$ . To allow the non-linear inviscid structures to fully develop, a very low kinematic viscosity is set ( $\nu = 10^{-6}$ ). The system is advanced up to  $t = 20$  s with a time step  $\Delta t = 10^{-3}$  s.

### 5.2 Results and Discussion

#### Qualitative validation

The physical evolution of the Kelvin-Helmholtz instability is governed by the highly non-linear coupling between the kinematic field and the pressure field. To visually assess this relationship, Figure 7 presents a side-by-side comparison of the velocity magnitude (with seeded directional glyphs) and the static pressure fields at four critical stages of the simulation:  $t = 1.3, 5.0, 11.5$ , and 17.0 seconds.

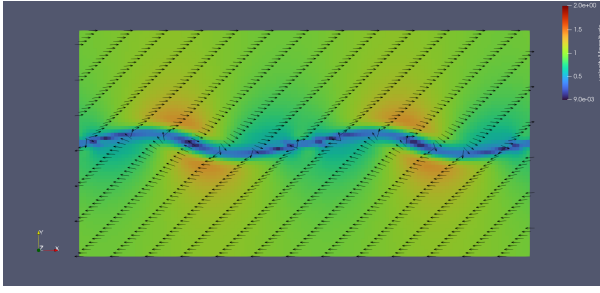
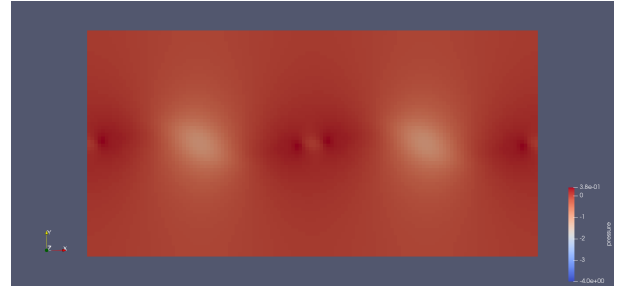
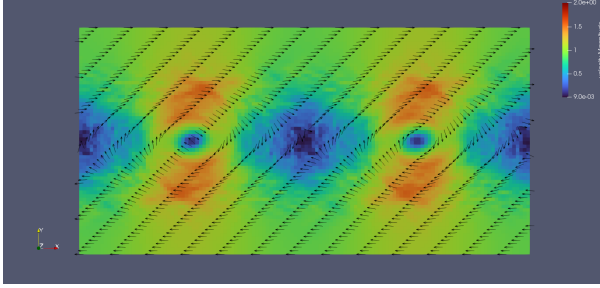
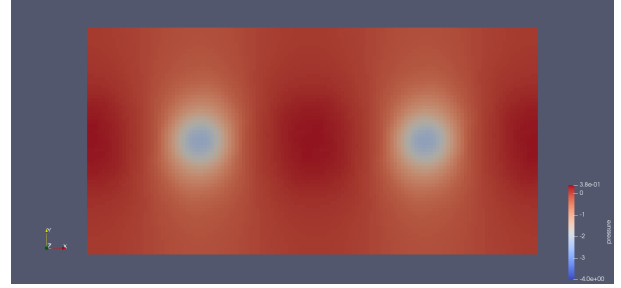
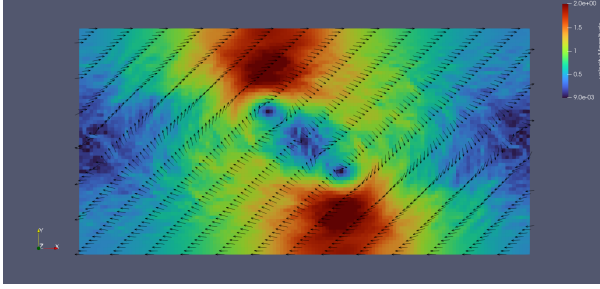
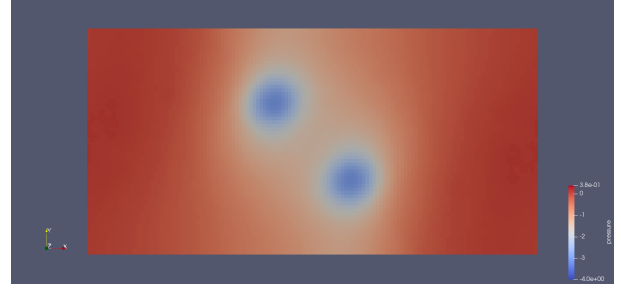
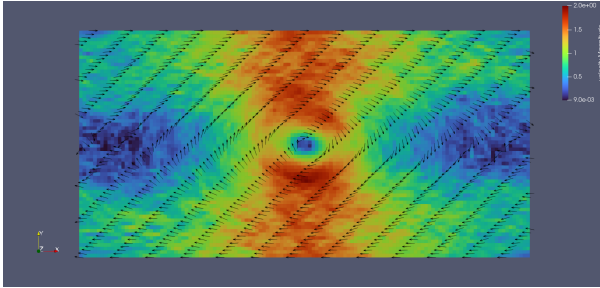
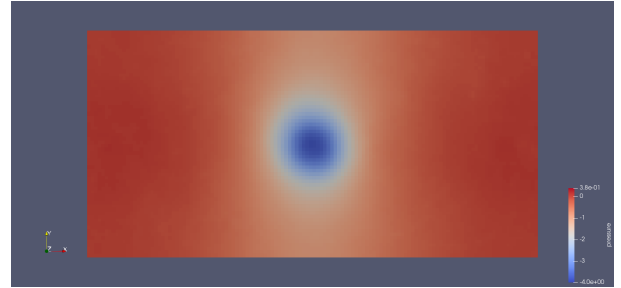
(a) Velocity field,  $t = 1.3$  s(b) Pressure field,  $t = 1.3$  s(c) Velocity field,  $t = 5.0$  s(d) Pressure field,  $t = 5.0$  s(e) Velocity field,  $t = 11.5$  s(f) Pressure field,  $t = 11.5$  s(g) Velocity field,  $t = 17.0$  s(h) Pressure field,  $t = 17.0$  s

Figure 7: Side-by-side evolution of the Kelvin-Helmholtz instability. The left column displays the velocity magnitude overlaid with directional vectors, while the right column shows the corresponding pressure distribution.

During the linear growth phase (for  $t \leq 1.3s$ ) (figures 7a and 7b), the initial vertical perturbation amplifies, causing the shear layer to undulate. The velocity vectors clearly indicate the formation of a wavy interface, while the pressure field exhibits weak, alternating high and low-pressure bands along the centerline, corresponding to the wave's crests and troughs.

As the flow transitions into the highly non-linear regime, the shear layer undergoes a roll-up process. The velocity discontinuity forces the fluid to fold onto itself, generating two distinct, tightly wound macroscopic vortices (7c). In strictly incompressible flows, the centripetal force required to sustain such intense rotational motion must be provided by a strong radial pressure gradient (7d). This physical requirement is perfectly captured by the numerical solver, as evident in the

pressure field at  $t = 11.5$  s: two deep, localized pressure minima (dark blue regions) emerge exactly at the geometric centers of the newly formed velocity vortices. Furthermore, the regions between the vortices (the braids) act as dynamic stagnation zones, characterized by distinct high-pressure peaks.

Around  $t = 11$  s the phenomenon of vortex pairing starts. The two small vortices start to orbit around each other and when sufficiently near they merge in a single bigger vortex (7g). A more detailed view of the co-rotation of the two vortices is shown as a sequence in figure 8.

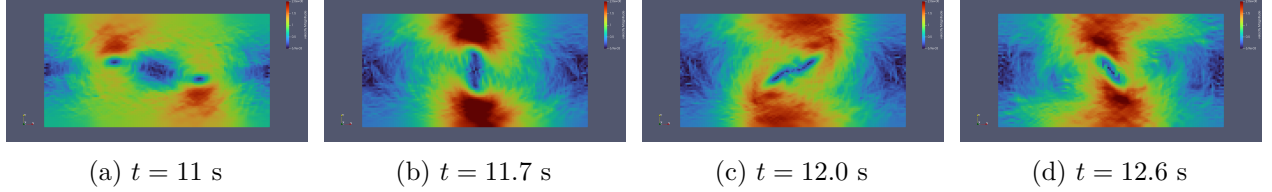


Figure 8: Temporal evolution of the spanwise vorticity field  $\omega_z$  during the vortex pairing phase: (a) two co-rotating vortices, (b)–(c) onset of mutual induction and orbital motion, (d) final merged state

## Quantitative analysis

Unlike the Taylor-Green vortex benchmark, the highly non-linear topological transition and the subharmonic interactions characterizing the Kelvin-Helmholtz instability lack a closed-form analytical exact solution. Consequently, the quantitative validation of the solver focuses on macroscopic flow statistics, the strict enforcement of physical constraints, and the robustness of the fractional-step method under extreme shear conditions.

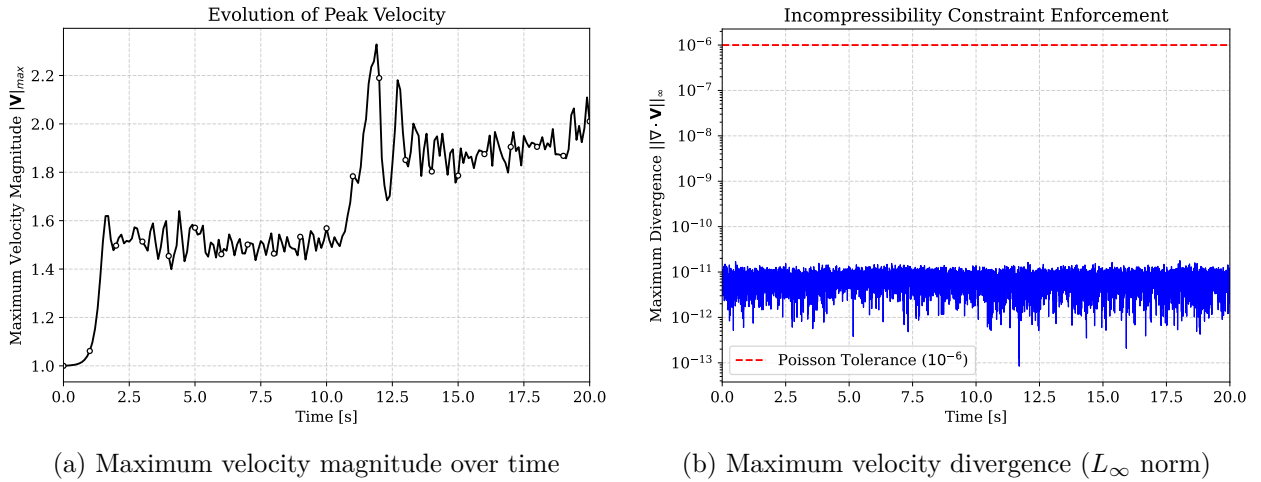


Figure 9: Global quantitative metrics of the simulation. Left: evolution of the velocity peak due to vortex induction. Right: rigorous enforcement of the incompressibility constraint.

The temporal evolution of the maximum velocity magnitude, extracted directly from the computational grid, is reported in Figure 9a. Initially, the peak velocity is bounded by the Dirichlet boundary conditions applied at the moving walls ( $|\mathbf{V}| \approx 1.0$ ). However, as the shear layer rolls up and the macroscopic vortices form ( $t \approx 1.5 - 11$  s), the mutual induction described by the Biot-Savart law locally accelerates the fluid particles along the outer edges of the vortex cores, leading to transient peaks that exceed the baseline flow velocity.

To assess the robustness of the projection method in handling these violent non-linear accelerations, Figure 9b tracks the maximum divergence of the corrected velocity field (`div_corr_max`). The theoretical analytical value for an incompressible fluid is strictly zero. The data confirms that throughout the entire simulation—including the critical pairing phase—the divergence remains bounded near machine-epsilon levels, strictly dictated by the Poisson solver tolerance set to  $10^{-6}$ .

This proves that the fractional-step architecture does not introduce artificial compressibility, even in the presence of nearly singular velocity gradients ( $\nu = 10^{-6}$ ).

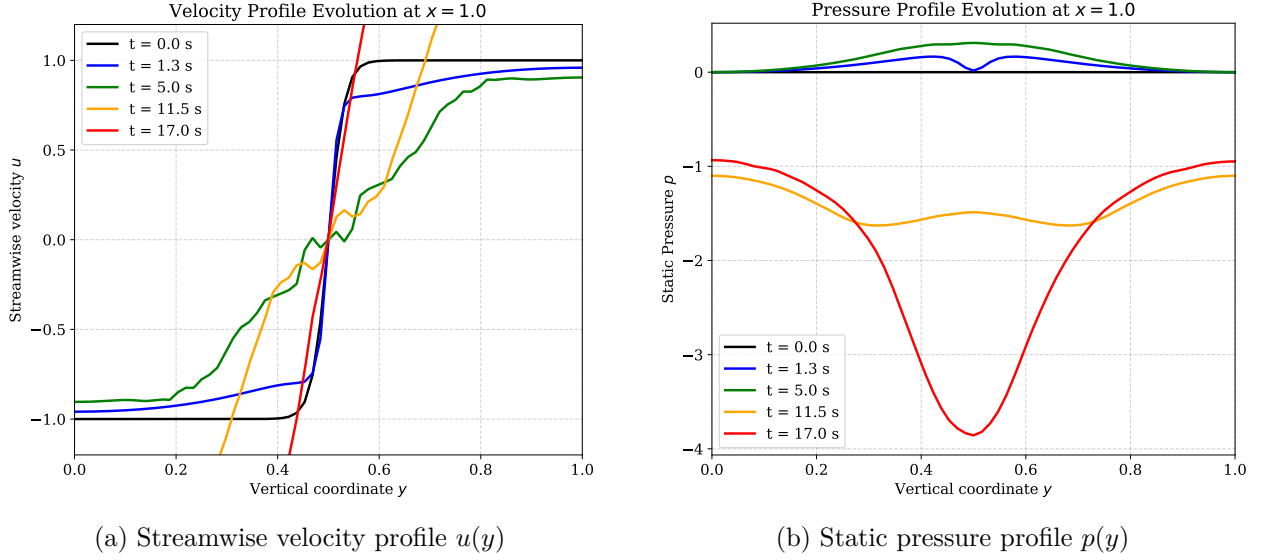


Figure 10: Evolution of the kinematic and dynamic profiles along the vertical centerline ( $x = 1.0$ ). (a) Progressive spreading of the shear layer due to momentum mixing. (b) Severe local pressure drops induced by the passage of the macroscopic vortex cores.

Finally, to quantify the cross-stream momentum transfer and the dynamic footprint of the vortices, Figure 10 plots the streamwise velocity  $u$  and the static pressure  $p$  extracted along the vertical centerline ( $x = 1.0$ ).

Regarding the kinematic behavior (Figure 10a), the profile at  $t = 0$  s perfectly matches the analytical hyperbolic tangent initialization. As time progresses, the vortex roll-up entrains external fluid into the core, translating into a progressive smoothing of the velocity gradient and a widening of the momentum thickness. Furthermore, the non-linear vortex dynamics perfectly explain the specific shapes of the advanced profiles ( $t = 11.5$  s and  $t = 17.0$  s). The velocity overshoots exceeding the freestream boundaries ( $|u| > 1.0$ ) are physically driven by the strong mutual induction of the vortices. Specifically, the steep, nearly linear central segment at  $t = 17.0$  s is the classic kinematic signature of a Rankine vortex core undergoing solid-body rotation, while the distinct inflections at  $t = 11.5$  s capture the complex, transient topology of the two unmerged cores crossing the extraction line during the subharmonic pairing phase.

Simultaneously, the static pressure profiles (Figure 10b) provide a striking quantitative confirmation of the vortex dynamics. While the initial pressure field is essentially uniform and close to zero, the passage of the macroscopic vortical structures through the extraction line induces severe, highly localized pressure drops. Specifically, during the pairing and merging phases ( $t = 11.5$  s and  $t = 17.0$  s), the pressure at the domain centerline ( $y = 0.5$ ) plummets to profound negative values, peaking near  $-4.0$ . This intense suction peak represents the strict dynamic requirement necessary to balance the extreme centrifugal forces generated by the rapidly spinning fluid within the core of the newly merged macro-vortex.

While the previous sections demonstrated the accuracy of the solver in capturing complex non-linear physics and strictly enforcing incompressibility, it is necessary to briefly comment on the method's sensitivity to the spatio-temporal resolution.

The fractional-step method, integrated in time via the explicit Runge-Kutta 3 (RK3) scheme, is conditionally stable. The chosen time step ( $\Delta t = 10^{-3}$  s) on the current  $128 \times 64$  spatial mesh guarantees that the advective Courant-Friedrichs-Lewy (CFL) number remains strictly below the theoretical stability threshold throughout the entire simulation. Even during the violent vortex pairing phase, where mutual induction accelerates local fluid particles beyond the initial unperturbed

velocity ( $|\mathbf{V}| > 1.0$ ), the fine temporal resolution successfully prevents the onset of unbounded numerical instabilities.

Regarding spatial sensitivity, the employed central difference scheme perfectly captures the inviscid roll-up without injecting excessive numerical dissipation. However, as the mixing layer grows and the connecting vortex filaments (the braids) are stretched to infinitesimally thin sheets, the finite grid resolution inherently acts as a spatial low-pass filter. While the current mesh ( $dx \approx dy \approx 0.015$ ) successfully resolves the primary macro-structures and the fundamental subharmonic pairing, capturing secondary, finer-scale Kelvin-Helmholtz instabilities along the braids would strictly require a systematic grid refinement study. Nevertheless, the complete absence of pressure checkerboard anomalies confirms the exceptional robustness of the implemented numerical architecture.

## 6 Conclusions

The developed Fractional-Step RK3 solver demonstrated excellent stability, strict enforcement of the incompressibility constraint, and formal second-order spatial accuracy.

- Through the Poiseuille flow validation, the code proved its reliability in handling wall-bounded viscous flows, accurately capturing the boundary layer development and the core mass-conservation acceleration.
- The Taylor-Green vortex benchmark further verified the robustness of the central-difference discretization, showcasing perfect preservation of complex flow topologies and highlighting the subtle interplay between physical viscosity and numerical dissipation.
- The unforced Kelvin-Helmholtz free simulation successfully validated the solver's capability to resolve highly non-linear, transient aerodynamic phenomena. The numerical architecture flawlessly captured the macroscopic vortex roll-up, the subharmonic pairing interactions, and the profound localized pressure drops required by the centripetal dynamics, maintaining the velocity divergence securely bounded at machine-epsilon levels even in under-resolved, nearly inviscid regimes.